

# Juan Esteban Flórez-Coronel

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## Education

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| <b>Ph.D. in Mechanical Engineering – GPA: 4.00</b><br><i>Texas A&amp;M University</i><br><i>Coursework: Numerical Methods for PDEs, Finite Elements, Material Science, Design Optimization</i><br><i>Research Focus: Bayesian optimization, Accelerated material discovery, High Performance Computing</i>                                               | Aug. 2024 – Present<br>College Station, TX |
| <b>M.S. in Scientific Computing – GPA: 3.60</b><br><i>University of Puerto Rico</i><br><i>Coursework: Numerical Analysis, Numerical Linear Algebra, Bayesian Machine Learning, Data Mining, Automatas and Formal Languages</i><br><i>Thesis: A Bayesian Adaptive Smoothing and Thresholding Approach for Activation Detection in Single-Subject fMRI</i> | Aug. 2021 – July 2024<br>Mayagüez, PR      |
| <b>Data Science Certification</b><br><i>Henry School</i><br><i>Coursework: Data Engineering, Data Analytics, Machine Learning</i>                                                                                                                                                                                                                        | May. 2023 – Dec. 2023<br>Argentina         |
| <b>B.S. in Mechanical Engineering – GPA: 3.91</b><br><i>University of Puerto Rico</i><br><i>Coursework: Adv. Thermodynamics, Computing Founds, Modern Algebra, Engineering Optimization, Funds of Mathematics, Graph Theory, Logic</i>                                                                                                                   | Aug. 2017 – May 2021<br>Mayagüez, PR       |

## Experience

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| <b>Graduate Teaching Assistant</b><br><i>MSEN655/489: Materials Design Studio</i>                             | Jan. 2026 – Present<br>Texas A&M University                           |
| <b>Graduate Research Assistant</b><br><i>Computational Design Laboratory</i>                                  | Aug. 2024 – Present<br>Texas A&M University                           |
| <b>Sports Statistician</b><br><i>Semi-Pro Rank: Football, Volleyball, Soccer, Basketball, Ice Hockey</i>      | Jan. 2024 – May 2025<br>Genius Sports                                 |
| <b>Deputy Leader: Team Puerto Rico</b><br><i>International Mathematical Olympiad</i>                          | July 2024<br>United Kingdom                                           |
| <b>Data Scientist</b><br><i>Caribbean Climate Adaptation Network</i>                                          | Jan. 2024 – July 2024<br>University of Puerto Rico: Medical Sciences  |
| <b>Developer and Administrator of Pre-Calculus Digital Book</b><br><i>Department of Mathematical Sciences</i> | Aug. 2023 – July 2024<br>University of Puerto Rico: Mayagüez          |
| <b>Graduate Research Assistant</b><br><i>HASPNEL Project</i>                                                  | Jan. 2023 – Dec. 2023<br>University of Puerto Rico: Mayagüez          |
| <b>Graduate Teaching Assistant</b><br><i>Department of Mathematical Sciences</i>                              | Aug. 2021 – Dec. 2022<br>University of Puerto Rico: Mayagüez          |
| <b>Mathematics Support Center Tutor</b><br><i>Department of Mathematical Sciences</i>                         | Aug. 2021 – Dec. 2023<br>University of Puerto Rico: Mayagüez          |
| <b>Biostatistician</b><br><i>NIH All of Us Program</i>                                                        | June. 2022 – July 2022<br>University of Puerto Rico: Medical Sciences |
| <b>Manufacturing Laboratory Assistant</b><br><i>Department of Mechanical Engineering</i>                      | Aug. 2021 – May 2022<br>University of Puerto Rico: Mayagüez           |
| <b>Data Analyst Programmer</b><br><i>National Institute of Congestion Reduction</i>                           | Aug. 2020 – Dec. 2020<br>University of Puerto Rico: Mayagüez          |

## Skills

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| <b>OS:</b> Linux, ChromeOS, iOS, MacOS Windows, Raspbian                                                                                     |
| <b>Tools:</b> AWS, Google Cloud, Local Servers (SSH), Google Workspace Tools, Microsoft Office Tools, Hadoop, Spark, Power BI, DAX, $\LaTeX$ |
| <b>Programming Languages:</b> Python, MATLAB, Arduino, R, JAVA, C, C++, Fortran, HTML, CSS, SQL, CUDA, MPI                                   |
| <b>CAD, CAM, FEA, CFD &amp; Others:</b> ANSYS, Star-CCM+, Siemens NX, SolidWorks, OnShape, Inventor, OpenRocket                              |
| <b>Manufacture:</b> 3D Printing, CNC Cutting Machines, Lathe, Milling and other mechanical machines                                          |
| <b>Project Management:</b> Gantt Charts, MeisterTask, Risk Assesments, Trello, Work Breakdown Structures (WBS)                               |
| <b>Languages:</b> Spanish (native), English (fluent)                                                                                         |

**ExactBO: GPU-Accelerated Certified Bayesian Optimization for Continuous and Constrained Materials Design**

*Other Authors: Raymundo Arróyave, Douglas Allaire*

Accepted: Conference Presentation 2027 TMS Annual Meeting

**Uncertainty-Propagating Active Learning Accelerates High-Entropy Oxide Discovery via Pulsed Electrodeposition for Hydrogen Evolution**

*Other Authors: Shakti Padhy, Neha Mishra, Subhajit Das, Ankur Srivastava, Doğuhan Sarıtürk, Suhash Dey, Douglas Allaire, Raymundo Arróyave*

Accepted: Conference Presentation 2027 TMS Annual Meeting

**An Automated Probabilistic Machine Learning Framework for Materials Modeling and Active Learning: Benchmarking Uncertainty Quantification Across Analytical, Computational, and Experimental Domains**

*Other Authors: Shakti Padhy, Sushant Sinha, Sk Md Ahnaf Akif Alvi, Robert Robinson, Jatin Ralhan, Raj Kiran, Douglas Allaire, Raymundo Arróyave*

Accepted: Conference Presentation 2027 TMS Annual Meeting

**Agentic Resource Allocation for Batch Multi-Objective Bayesian Optimization in Autonomous Materials Discovery**

*Other Authors: Robert Robinson, Shakti Padhy, Sushant Sinha, Ahnaf Alvi, Brent Vela, Trevor Hastings, Douglas Allaire, Raymundo Arróyave*

Accepted: Conference Poster 2027 TMS Annual Meeting

**Mechanics-Based Multidisciplinary Design Optimization of Shape-Memory Hook Fasteners for Programmable Interlocking**

*Other Authors: Maria Vallejo-Ciro, Manuel Carvajal-Loaiza, Mohamad El Halabi, Rahaf Homssi, Vanessa Restrepo*

Conference Paper ASME SMASIS2026-190820, DOI: Coming Soon

**ExactBO: GPU-Accelerated Interval Arithmetic for Exact Expected-Improvement Maximization in Bayesian Optimization**

*Other Authors: Raymundo Arróyave, Douglas Allaire*

Conference Paper IDETC2026-187247, DOI: Coming Soon

**Mapping of microstructure transitions during rapid alloy solidification using Bayesian-guided phase-field simulations**

*Other Authors: José Mancias, Brent Vela, Rouhollah Tavakoli, Douglas Allaire, Raymundo Arróyave, Damien Tournet*

Paper DOI: <https://doi.org/10.1016/j.actamat.2025.121354>

**Incorporating Equity in the Vehicle Rebalancing Operations of Dockless Micromobility Services**

*Other Authors: Lina Zapata, Daniel Rodriguez-Roman, Juan González-López and Alberto Figueroa-Medina*

Paper DOI: <https://doi.org/10.1016/j.latran.2024.100009>

**Bayesian Adaptive Smoothing for Activation Detection in fMRI**

*Advisor: Israel Almodóvar*

Conference Presentation in 2023 SACNAS NDiSTEM Conference and 2023 BEER-XIV Conference

**Equity Considerations in the Rebalancing Operations of Dockless Micromobility**

*Other authors: Lina Zapata, Daniel Rodriguez-Roman, Juan González-López and Alberto Figueroa-Medina*

Conference Paper from XX PANAM 2021 hosted by the Pan American Society of Transportation Research

**Prediction of Trip Productions for Exclusive Zones of a Dockless Electric Scooter Service using Machine Learning Regression Algorithms**

*Other authors: Daniel Rodriguez-Roman and Benjamín Colucci-Ríos*

Conference Poster for 2020 Transportation Research Board Annual Meeting

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Projects

**BIRDSHOT**

*High-Throughput Materials Discovery for Extreme Environments Center*

Jan. 2025 – Present

*Texas A&M University*

- Assembled and operated the Multi-Objective Batch Bayesian Optimization (MOBBO) framework for the iterative discovery of alloys for extreme environments, using Gaussian process surrogate models and expected hypervolume improvement to balance competing material properties
- Select 12 of the 24 alloy compositions advanced during each design–make–test–learn iteration by evaluating model predictions, uncertainty, feasibility constraints and expected experimental value
- Present and defend alloy recommendations to an interdisciplinary team of computational researchers, materials scientists and experimentalists, informing the allocation of approximately \$20,000 in synthesis, processing, characterization and testing per iteration

**DRACO Rocketry Team**

*NASA Student Launch University Initiative (NSL)*

Aug. 2018 – May 2021

*University of Puerto Rico: Mayagüez*

- Led a team in the design, manufacture and launch of a high-power rocket with an experimental payload to an approximate height of 4,500 ft, recover it and complete a specified mission
- Developed mathematical models and computational simulations in order to understand and predict the mission performance of the rocket with less than 3% of error
- Managed the project using Systems Engineering tools such as requirement establishment, risk assessments, decision matrix, single-point failures, BOM by order of assembly, proposal, PDR, CDR, FRR and professional presentations, all with high quality as awarded by NASA panel

## Workshops

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### **Advanced Computing Workshop**

*Institute of Data Science*

### **Computational Materials Science Summer School**

*Department of Materials Science and Engineering*

### **Sampling Advanced Mathematics for Minority Students Program**

*Department of Mathematics*

August 2025

*Texas A&M University*

July 2025 – Aug. 2025

*Texas A&M University*

June 2019 – July 2019

*Ohio State University*

## Awards

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- **Finalist** ASME SMASIS 2026 Best Student Paper Award
- **Best Presentation** SACNAS NDiSTEM Conference 2023
- **First Place** Programming Competition UPR-Bayamon 2022
- **Fellowship** Dwight David Eisenhower Transportation 2020
- **First Place** Project Review in NSL 2020
- **Rookie Award** NSL 2020
- **Second Place** STEM Engagement in NSL 2020
- **Top 25% and 30%** Putnam Math Competition 2019 and 2018
- **First Place** Calculus Olympiads UPRM 2018
- **Third Place** Calculus Olympiads Interamerican University 2018
- **Gold Medal** OMPR Second and Third Phases 2015 - 2017